

DS4200 Project Description

Project website link: <https://liyaliju.github.io/StudentPerformanceDataVizProject/>

Notes to the grading TA: –

Design ideas:

1. Sleep Hours and GPA Change

This visualization is a scatter plot made with Altair that shows the relationship between the number of hours a student sleeps per night and their change in GPA from the previous semester to the current one. The mark we chose was points, and the primary channels we used were horizontal position for sleep hours, vertical position for GPA change, and color to distinguish between technical and non-technical majors. We added a dashed reference line at $y = 0$ to make it immediately clear which students improved versus declined, so the audience does not have to interpret the axis values to understand the direction of change. We also included three interactive features: a radio button to filter by major type, an age slider to filter by student age, and a brush selection on the scatter plot that links to a bar chart below showing the count of students per major in the selected region.

2. Overview of Student Habits by Major

Our second visualization is a stacked bar chart made with D3 that shows the average time spent studying, sleeping, and socializing by major and major type. We place it after the visualizations showing sleep hours and GPA change by major so that the audience knows all of the different majors included and which majors fall under technical and non-technical. The mark we chose was areas, and the primary channels we used were color and vertical position to distinguish between each category of student habits and encode the number of hours spent for each one. We chose to include an interactive tooltip for an exact reading of the number values.

3. Correlation Between Habits and Performance

The heatmap made with D3 shows the correlation between student habits, attendance rate, number of hours of sleep, number of study hours, and number of study hours, and the student's final GPA grouped by major and age. One design choice was picking the color scheme to be from red to yellow to green to represent negative correlation to no correlation to positive correlation. We also included the Pearson correlation on each cell so the user could clearly see the actual correlation numbers. We also added a toggle between grouping by ages and by majors to let the user see two different ways the data can be organized. Finally, we also included

an interactive tooltip feature that included the habit, either the age or major, the correlation coefficient, and how strong or weak it is correlated with GPA.

4. Study Hours Per Day by Major

This visualization is a boxplot made with Altair that shows the distribution of daily study hours across six majors. The mark we chose was boxes, and the primary channels we used were horizontal position for study hours and vertical position for major. We chose a single consistent color for all majors rather than encoding major by color, since color would be redundant given that each major is already labeled on the y-axis. The box and whisker format was chosen over a bar chart because it conveys the full spread of the data, including the median, interquartile range, and outliers, giving the audience a more complete picture of how study habits vary within each major rather than just the average. We also added an interactive tooltip so the audience can highlight a specific major to compare it more easily against the others.

5. Study Hours Per Day vs GPA Change

This visualization is a scatter plot made with Altair that shows the relationship between the number of hours a student studies per day and their change in GPA. The mark we chose was points, and the primary channels we used were horizontal position for study hours and vertical position for GPA change. We chose a single color for all points since encoding by major or major type did not reveal any meaningful separation between groups. We added a linear regression trend line to make the overall positive relationship more explicit and easier for the audience to interpret at a glance, and a dashed reference line at $y = 0$ to clearly separate students who improved from those who declined. A tooltip was also included so the audience can hover over individual points to see the student's major, study hours, and GPA change.

6. Attendance and Study Hours vs GPA Change

For this visualization, we wanted to see how each individual student was doing when regarding their attendance and study hours, and see if their GPA change had trends regarding both of these statistics. We decided to use color as a mark for Study Hours, as it is easily discernible when taking a quick look at the graph to find a pattern, and put the Attendance and GPA change on the X and Y axis because we thought that these two metrics would be significant when looking at Study Hours, both positive and negative. Color also has the advantage of being able to be a two color palette, where we can use a color like Red to mark a lower amount of study hours, yellow to be a middling amount of study hours, and Green to be a very large amount of study hours, as we did in this graph, and it has made it very easy to see trends because of this change we made. By putting attendance on the X axis, and GPA change on the Y axis, it also makes it easier for us to spot outliers that could be contributing to a smaller GPA change, or

even a negative change, even though they are spending a lot of time studying for a class. We also added a trend line to easily identify the general trend the data is showing

7. Gender and Number of Students in Major

For this visualization, we wanted to make sure that we could categorize not only just by gender or major, but break down each of them into their own subcategories, and then rank them by attendance percentage. To do this, we created an X axis where Bars in our bar chart are broken down by Gender, and then Technical vs Non Technical Majors, and then finally the actual major division itself. By doing this, we compare each Major within the realm of the student body that it relates to, and so we are able to make accurate comparisons that would be relevant to each major regarding attendance. The reason we have chosen to go with a Bar Chart instead of a different multi-axis chart, which could have been another way to visualize this information is due to the Bar chart having color be a mark, alongside the Y axis being the attendance percentage, so we can easily compare student count alongside attendance percentage, as both are variables that can affect the other.

8. Gender and Major GPA Change Sankey

For this visualization, we wanted to make sure that we could see the changes of GPA grouped by the categories that we are interested in. We chose Gender and Major as our two categories where we wanted to take a look at exactly how each of these specific groups have their GPA being changed. Since we are using Tableau, the chart is interactive, and so when you want to see each grouping's GPA change, we have an interaction where you can hover over the group, and it shows how much the change is. We have also added percentages on the Sankey groupings themselves, so you can see how much of the GPA changes are being done by Gender, as well as by Major. In the tooltip itself for each combination between Major and Gender, we have added a Delta of GPA change compared to the overall average GPA change, to easily compare each Sankey connection to the overall set of students.

Feedback from the peer review:

The main feedback we got from the peer review is that one of our bar graphs showing attendance rate by major did not meaningfully add to the website, so we replaced the graph by splitting one of our other visualizations into two distinct graphs: one to show attendance vs. GPA change and one to show study hours vs GPA change. We also received feedback about adding a regression line to our scatterplots and removing color from the Study Hours Per Day by Major boxplot, so we added a regression line to the scatterplots we made and removed color as an encoding channel.

The changes we made include rearranging the order of the visualizations to tell a clearer story with better flow for the audience, sorting the majors in a consistent order across all visualizations, replacing one of our bar graph visualizations with a scatter plot, adding interaction instructions and tool tags indicating whether each visualization was made with Altair, D3, or Tableau, and redesigning the website to enhance its styling and professional appearance with easy navigation to each visualization.

The reason we did not change the color scale used for the heatmap, even though we received feedback to pick a different color scale, is because the diverging traffic light scale (Red-Yellow-Green) we are using aligns with the message we are communicating to the audience. We want to show that high positive numbers are “good”, high negative numbers are “bad”, and any numbers in the yellow range (i.e. low positive or negative numbers close to zero) are neutral, and not necessarily leaning in either “good” or “bad” direction.